

★ MONTH 5 – COMPLETE REMAINING CLASS 11 (JEE FOUNDATION STRONG)

■ PHYSICS (Class 11)

1. Rotational Motion

Subtopics:

1. Rigid Body Basics
 2. Moment of Inertia (MOI)
 - Rod, Disc, Ring, Sphere, Cylinder
 3. Parallel & Perpendicular Axis Theorem
 4. Angular Velocity & Acceleration
 5. Rolling Motion
 6. Torque & Angular Momentum
 7. Conservation of Angular Momentum
 8. Rotational Kinetic Energy
 9. Pure Rolling (Very important for JEE)
 10. COM (Centre of Mass)
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2. Gravitation

Subtopics:

1. Newton's Law of Gravitation
 2. Gravitational Field & Potential
 3. Escape Velocity
 4. Orbital Velocity
 5. Kepler's Laws
 6. Satellite Motion
 7. Variation of g with
 - Height
 - Depth
 - Rotation of Earth
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3. SHM (Simple Harmonic Motion)

Subtopics:

1. SHM Definition
 2. Equation of SHM
 3. Energy in SHM
 4. Spring-Mass System
 5. Pendulums (Simple pendulum & physical pendulum)
 6. Damped & Forced Oscillations (conceptual)
 7. Differential Equation of SHM
 8. Phase, Displacement, Velocity Graphs
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4. Waves

Subtopics:

1. Types of Waves (Transverse, Longitudinal)
 2. Wave Equation
 3. Speed of Wave in String
 4. Principle of Superposition
 5. Interference
 6. Beats
 7. Doppler Effect
 8. Standing Waves
 9. Organ Pipes → Open & Closed
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CHEMISTRY (Class 11)

1. GOC + Organic Chemistry Basics

Subtopics:

1. Classification of Organic Compounds
2. IUPAC Nomenclature
3. Isomerism (Structural + Stereoisomerism basics)
4. Inductive Effect
5. Mesomeric Effect
6. Hyperconjugation
7. Carbocations, Carbanions, Free Radicals
8. Resonance
9. Acidic & Basic Strength

10. Reaction Mechanisms basics

- SN1
 - SN2
 - E1
 - E2
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2. Hydrocarbons

Subtopics:

Alkanes

1. Preparation
2. Properties
3. Free Radical Substitution

Alkenes

4. Preparation
5. Markovnikov Rule
6. Addition Reactions

Alkynes

7. Acidic Nature
8. Important Reactions

Aromatic Hydrocarbons

9. Benzene → Electrophilic Substitution
 10. Directive Influence (ortho/para/meta)
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3. Thermodynamics

Subtopics:

1. System & Surroundings
2. Work, Heat, Internal Energy
3. First Law of Thermodynamics
4. Enthalpy
5. Enthalpy changes (combustion, formation, hydration)
6. Hess's Law
7. Bond Enthalpy
8. Spontaneity – Gibbs Free Energy
9. Entropy (very important for JEE)

4. Equilibrium

Subtopics:

Chemical Equilibrium

1. Law of Mass Action
2. Le Chatelier's Principle
3. Equilibrium Constant – K_p, K_c
4. Relationship between K_p & K_c
5. Applications to Reactions

Ionic Equilibrium

6. pH & pOH
 7. Buffer Solutions
 8. Solubility Product (K_{sp})
 9. Common Ion Effect
 10. Hydrolysis of Salts
 11. Acid–Base Theory (Arrhenius, Bronsted, Lewis)
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MATHEMATICS (Class 11)

1. Sequence & Series

Subtopics:

1. Arithmetic Progression (AP)
 2. Geometric Progression (GP)
 3. Mixed Series
 4. Sum of n terms
 5. AM, GM, HM
 6. Special Series
 - $1^2 + 2^2 + \dots$
 - $1^3 + 2^3 + \dots$
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2. Permutation & Combination (P&C)

Subtopics:

1. Fundamental Principle of Counting

2. Permutations
 3. Factorials
 4. Circular Permutations
 5. Combinations
 6. Pascal's Triangle
 7. Problem types (arrangements, selections)
 8. Restricted arrangements
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3. Coordinate Geometry

Subtopics:

Straight Line

1. Slope
2. Various forms of equations
3. Distance, Section formula
4. Angle between lines

Circle

5. Equation of circle
6. Tangent & Normal
7. Power of a Point

Parabola, Ellipse, Hyperbola (Basics)

8. Standard equations
 9. Focus, Directrix, Eccentricity
 10. Basic problems (important for JEE)
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MONTH 5 – 4-WEEK SCHEDULE

WEEK 1

Physics: Rotation

Chemistry: GOC + Organic Basics

Math: Sequence & Series

WEEK 2

Physics: Gravitation

Chemistry: Hydrocarbons

Math: Permutation & Combination

WEEK 3

Physics: SHM

Chemistry: Thermodynamics

Math: Coordinate Geometry (Lines + Circles)

WEEK 4

Physics: Waves

Chemistry: Equilibrium

Math: Coordinate Geometry (Conics)

Full PYQs + Mixed JEE Problems